American Computer Science League

2025-2026 ● Contest 2: Short Problems ● Elementary Division Solutions

1. Prefix/Infix/Postfix Notation

9 3 * 9 2 3 ^ − + 8 1 − /
= ((9 * 3) + (9 − 2 ^ 3)) / (8 − 1)
= (27 + (9 − 8)) / 7
= (27 + 1) / 7
= 28 / 7
= 4

C. 4

2. Prefix/Infix/Postfix Notation

/ * 8 − 9 4 + ^ 3 2 1
= (8 * (9-4)) / ((3^2) + 1)
= 40 / 10
= 4

A. 4

3. Prefix/Infix/Postfix Notation

$$\frac{4+6^2-5}{8}+7\cdot 2$$

= (4 + 6 ^ 2 − 5) / 8 + 7 * 2
= (4 + (^ 6 2) − 5) / 8 + (* 7 2)
= ((+ 4 (^ 6 2)) − 5) / 8 + (* 7 2)
= (− (+ 4 (^ 6 2)) 5) / 8 + (* 7 2)
= (/ (− (+ 4 (^ 6 2)) 5) 8) + (* 7 2)
= + (/ (− (+ 4 (^ 6 2)) 5) 8) (* 7 2)
= + / − + 4 ^ 6 2 5 8 * 7 2

B. + / − + 4 ^ 6 2 5 8 * 7 2

4. Prefix/Infix/Postfix Notation

$(7 − 5)^2$ + 9 * (6 − 2) / 4
= (((7 5 −) ^ 2) + (((6 2 −) * 9) / 4))
= (((7 5 −) 2 ^) + ((9 (6 2 −) *) / 4))
= (((7 5 −) 2 ^) + ((9 (6 2 −) *) 4 /))
= (((7 5 −) 2 ^) ((9 (6 2 −) *) 4 /) +)
= 7 5 − 2 ^ 9 6 2 − * 4 / +

D. 7 5 − 2 ^ 9 6 2 − * 4 / +

5. Prefix/Infix/Postfix Notation a. / − 8 4 2 = (8 − 4) / 2 = 2 b. − / 8 4 2 = (8 / 4) − 2 = 0 c. − 8 / 4 2 = 8 − (4 / 2) = 6 d. / 8 − 4 2 = 8 / (4 − 2) = 4 The largest to the smallest is c,d,a,b.	C. c, d, a, b
6. Prefix/Infix/Postfix Notation 5 3 * 6 2 / + 2 3 ^ − 5 / = ((((* 5 3) (/ 6 2) +) (^ 2 3) −) 5 /) = (((+ (* 5 3) (/ 6 2)) (^ 2 3) −) 5 /) = ((−1 (+ (* 5 3) (/ 6 2)) (^ 2 3)) 5 /) = (/ (− (+ (* 5 3) (/ 6 2)) (^ 2 3)) 5) = / − + * 5 3 / 6 2 ^ 2 3 5	E. / − + * 5 3 / 6 2 ^ 2 3 5